

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 13: Section and Mid-Point Formula

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Centroid of a triangle

1. Triangle ABC has vertices $A(0, 6)$, $B(4, -4)$ and $C(-4, -2)$. D is the point where AC crosses the x -axis. (a) Find the coordinates of D . (b) Find the centroid of $\triangle ABC$. (c) D divides AC in the ratio $k : 1$; find k . (d) Find the equation of the line BD . **ICSE 2024, 4 marks**

SOLUTION

(c) Let D divide AC in the ratio $k : 1$. Its y -coordinate is 0: $\frac{-2k + 6}{k + 1} = 0$, so $k = 3$.

(a) Then $x = \frac{3(-4) + 0}{3 + 1} = -3$, so $D = (-3, 0)$.

(b) Centroid = $\left(\frac{0 + 4 - 4}{3}, \frac{6 - 4 - 2}{3}\right) = (0, 0)$

(d) Slope of $BD = \frac{0 - (-4)}{-3 - 4} = -\frac{4}{7}$. Through D : $y = -\frac{4}{7}(x + 3)$, so $7y = -4x - 12$.

Answer: (a) $D(-3, 0)$ (b) $(0, 0)$ (c) $k = 3$ (d) $4x + 7y + 12 = 0$

2. The vertices of $\triangle ABC$ are $(-4, -2)$, $(6, 2)$ and $(4, 6)$. Find its centroid. (a) $(2, 2)$ (b) $(2, 3)$ (c) $(3, 3)$ (d) $(0, -1)$ **ICSE 2023, 1 mark**

SOLUTION

Centroid = $\left(\frac{-4 + 6 + 4}{3}, \frac{-2 + 2 + 6}{3}\right) = (2, 2)$

Answer: (a)

2 The section formula

3. $A(4, 0)$ lies on the x -axis and $B(0, 4)$ lies on the y -axis. P is the point on AB with $AP : PB = 3 : 1$. (a) Use the section formula to find P . (b) Find the equation of the line through P perpendicular to AB . **ICSE 2023, 4 marks**

SOLUTION

(a) $P = \left(\frac{3 \times 0 + 1 \times 4}{3 + 1}, \frac{3 \times 4 + 1 \times 0}{3 + 1}\right) = (1, 3)$

(b) Slope of $AB = \frac{4 - 0}{0 - 4} = -1$, so the perpendicular line has slope 1.

$y - 3 = 1(x - 1)$

Answer: (a) $P(1, 3)$ (b) $y = x + 2$

4. M lies on the x -axis and N lies on the y -axis. The point $P(3, 2)$ divides MN in the ratio $2 : 3$. Find (a) the coordinates of M and N (b) the slope of MN . **ICSE 2019, 3 marks**

SOLUTION

(a) Let $M = (a, 0)$ and $N = (0, b)$. Then $P = \left(\frac{2 \times 0 + 3a}{5}, \frac{2b + 3 \times 0}{5} \right)$.

$\frac{3a}{5} = 3$ gives $a = 5$, and $\frac{2b}{5} = 2$ gives $b = 5$.

(b) Slope of $MN = \frac{5 - 0}{0 - 5} = -1$

Answer: (a) $M(5, 0)$, $N(0, 5)$ (b) slope $= -1$

5. $P(1, -2)$ lies on the segment joining $A(3, -6)$ and $B(x, y)$, with $AP : PB = 2 : 3$. Find the coordinates of B . **ICSE 2017, 4 marks**

SOLUTION

$1 = \frac{2x + 3(3)}{5}$, so $2x = -4$ and $x = -2$.

$-2 = \frac{2y + 3(-6)}{5}$, so $2y = 8$ and $y = 4$.

Answer: $B(-2, 4)$

3 Finding the ratio

6. In what ratio does the y -axis divide the segment joining $P(5, 3)$ and $Q(-5, 3)$? Find the point of intersection. **ICSE 2020, 4 marks**

SOLUTION

Let the ratio be $k : 1$. On the y -axis the x -coordinate is 0: $\frac{-5k + 5}{k + 1} = 0$, so $k = 1$.

The point is the mid-point: $\left(0, \frac{3 + 3}{2} \right) = (0, 3)$.

Answer: ratio $1 : 1$; the point is $(0, 3)$

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